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Problem Statement

Analyze the supplied 2015 U.S. flight dataset using Hadoop MapReduce and Apache Pig. The analysis identifies flight volumes, airport activity, monthly demand, cancellation patterns, and delay performance using the fields available in the dataset..

Dataset Description

Dataset: <https://www.kaggle.com/datasets/usdot/flight-delays>

The dataset contains three CSV files: `flights.csv`, `airlines.csv`, and `airports.csv`.

The main file, `flights.csv`, contains **5,819,079 flight records** with **31 columns**. It includes information such as flight date, airline code, flight number, origin airport, destination airport, departure delay, arrival delay, cancellation status, cancellation reason, distance, and air time.

The `airlines.csv` file contains airline codes and airline names. The `airports.csv` file contains airport codes, airport names, cities, states, countries, latitude, and longitude. These two files are used as lookup datasets to understand airline and airport codes in the main flight data.

Field	Description
YEAR, MONTH, DAY	Flight date components
AIRLINE	Operating airline code
ORIGIN_AIRPORT, DESTINATION_AIRPORT	Departure and arrival airport identifiers
DEPARTURE_DELAY, ARRIVAL_DELAY	Delay in minutes
CANCELLED, CANCELLATION_REASON	Cancellation indicator and cause
DISTANCE, AIR_TIME	Route distance and airborne duration

How to obtain the dataset

- Download the 2015 Flight Delays and Cancellations dataset from Kaggle.
- Extract `flights.csv`, `airlines.csv`, and `airports.csv` into the project data folder.
- The files are comma-separated CSV files; `flights.csv` is the primary analysis input.
- Run the MapReduce and Pig commands from Windows Administrator Command Prompt.

Summary of Data Analysis

This project analyzes the airline flight dataset using Hadoop MapReduce and Apache Pig. The purpose of the analysis is to identify airline activity, busy airports, monthly flight trends, cancellation patterns, and flight-delay performance from a large collection of flight records.

Hadoop MapReduce is used to process the dataset through Driver, Mapper, and Reducer classes. The Mapper reads each flight record and extracts the required fields. Hadoop groups records with the same key, and the Reducer combines these values to generate the final results. Apache Pig is used for high-level analysis through Pig Latin scripts, which make it easier to load, filter, group, sort, and store the data.

- **Filtering:** Header records, invalid records, and incomplete flight records are excluded before analysis.
- **Data Organization:** Airline code, origin airport, month, and route fields are used as grouping keys to organize related flight records.
- **Summarization:** Reducers calculate total flight counts, airport departures, cancellation counts, and average delay values.
- **Performance Analysis:** Departure and arrival delay fields are analyzed to compare airline performance and identify airlines with higher delays.

MapReduce analyses performed:

- **Flight count by airline:** Counts the total number of flights operated by each airline.
- **Departures by origin airport:** Counts the number of flights departing from each airport.
- **Flight count by month:** Calculates the total number of flights for every month.
- **Cancelled flights by airline:** Counts cancelled flights for each airline.
- **Average arrival delay by airline:** Calculates the average arrival delay in minutes for each airline.

Pig analyses performed:

- **Airline flight totals:** Shows total flights for each airline with airline names.
- **Monthly flight totals:** Shows the total number of flights in each month.
- **Top origin airports:** Identifies the ten airports with the highest number of departures.
- **Top routes:** Identifies the ten busiest flight routes based on origin and destination airports.
- **Average departure delay by airline:** Calculates the average departure delay for each airline.
- **Cancellations by airline:** Shows the total number of cancelled flights for each airline.
- **Average arrival delay by month:** Calculates the average arrival delay for each month.

Map Reduce Analysis

1) Flight Count by Airline

Counts all scheduled flight records for each airline code.

Code

```
package flightanalysis;

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

/** Counts all scheduled flight records for each airline code. */
public class AirlineFlightCount {
    public static class Map extends Mapper<LongWritable, Text, Text, IntWritable> {
        private static final IntWritable ONE = new IntWritable(1);
        public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
            String[] f = value.toString().split(",", -1);
            if (f.length == 31 && !"YEAR".equals(f[0]) && !f[4].trim().isEmpty()) context.write(new Text(f[4]), ONE);
        }
    }
    public static class Reduce extends Reducer<Text, IntWritable, Text, IntWritable> {
        public void reduce(Text key, Iterable<IntWritable> values, Context context) throws IOException, InterruptedException {
            int total = 0; for (IntWritable value : values) total += value.get();
            context.write(key, new IntWritable(total));
        }
    }
    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration(); conf.set("fs.defaultFS", "file:///"); conf.set("mapreduce.framework.name", "local");
        Job job = Job.getInstance(conf, "Flight count by airline");
        job.setJarByClass(AirlineFlightCount.class); job.setMapperClass(Map.class); job.setReducerClass(Reduce.class);
        job.setOutputKeyClass(Text.class); job.setOutputValueClass(IntWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0])); FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Command to run

```
hadoop jar MapReduce\build\flightanalysis.jar flightanalysis.AirlineFlightCount data\flights.csv MapReduce\output\01_airline_flight_count
```

```
C:\Windows\System32>cd /d E:\BDAL\Final_Assignment\Flight_Data_Analysis

E:\BDAL\Final_Assignment\Flight_Data_Analysis>hadoop jar MapReduce\build\flightanalysis.jar flightanalysis.AirlineFlightCount data\flights.csv MapReduce\output\01_airline_flight_count
Map input records=5819080
Map output records=5819079
Reduce output records=14
map 100% reduce 100%
Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/MapReduce/01_airline_flight_count/part-r-00000
AA      725984
AS      172521
B6      267048
DL      875881
EV      571977
F9      90836
HA      76272
MQ      294632
NK      117379
OO      588353
UA      515723
US      198715
VX      61903
WN      1261855
```

Selected execution output

Key	Result
WN	1,261,855
DL	875,881
AA	725,984
OO	588,353
EV	571,977

2) Departures by Origin Airport

Counts flights departing from each origin airport.

Code

```
package flightanalysis;

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

/** Counts departing flights at each origin airport. */
public class AirportDepartureCount {
    public static class Map extends Mapper<LongWritable, Text, Text, IntWritable> {
        private static final IntWritable ONE = new IntWritable(1);
        public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
            String[] f = value.toString().split(", ", -1);
            if (f.length == 31 && !"YEAR".equals(f[0]) && !f[7].trim().isEmpty()) context.write(new Text(f[7]), ONE);
        }
    }

    public static class Reduce extends Reducer<Text, IntWritable, Text, IntWritable> {
        public void reduce(Text key, Iterable<IntWritable> values, Context context) throws IOException, InterruptedException {
            int total = 0; for (IntWritable value : values) total += value.get(); context.write(key, new IntWritable(total));
        }
    }

    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration(); conf.set("fs.defaultFS", "file:///"); conf.set("mapreduce.framework.name", "local");
        Job job = Job.getInstance(conf, "Departures by origin airport");
        job.setJarByClass(AirportDepartureCount.class); job.setMapperClass(Map.class); job.setReducerClass(Reduce.class);
        job.setOutputKeyClass(Text.class); job.setOutputValueClass(IntWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0])); FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Command to run

hadoop jar MapReduce\build\flightanalysis.jar flightanalysis.AirportDepartureCount data\flights.csv
MapReduce\output\02_airport_departure_count

```
C:\Windows\System32>cd /d E:\BDAL\Final_Assignment\Flight_Data_Analysis

E:\BDAL\Final_Assignment\Flight_Data_Analysis>hadoop jar MapReduce\build\flightanalysis.jar flightanalysis.AirportDepartureCount data\flights.csv MapReduce\output\02_airport_departure_count
Map input records=5819080
Reduce output records=628
map 100% reduce 100%
Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/MapReduce/02_airport_departure_count/part-r-00000 | powershell -Command "$input | Select-Object -First 15"
10135 239
10136 201
10140 1816
10141 71
10146 87
10154 28
10155 148
10157 122
10158 248
10165 9
10170 31
10185 282
10208 223
10257 734
10268 57
```

Selected execution output

Key	Result
ATL	346,836
ORD	285,884
DFW	239,551
DEN	196,055
LAX	194,673

3) Flight Count by Month

Aggregates flight records by calendar month.

Code

```
package flightanalysis;

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

/** Counts flight records in each calendar month. */
public class MonthlyFlightCount {
    public static class Map extends Mapper<LongWritable, Text, Text, IntWritable> {
        private static final IntWritable ONE = new IntWritable(1);
        public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
            String[] f = value.toString().split(", ", -1);
            if (f.length == 31 && !"YEAR".equals(f[0]) && !f[1].trim().isEmpty()) context.write(new Text(f[1]), ONE);
        }
    }
    public static class Reduce extends Reducer<Text, IntWritable, Text, IntWritable> {
```

```

    public void reduce(Text key, Iterable<IntWritable> values, Context context) throws IOException, InterruptedException {
        int total = 0; for (IntWritable value : values) total += value.get(); context.write(key, new IntWritable(total));
    }
}
public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration(); conf.set("fs.defaultFS", "file:///"); conf.set("mapreduce.framework.name", "local");
    Job job = Job.getInstance(conf, "Flight count by month");
    job.setJarByClass(MonthlyFlightCount.class); job.setMapperClass(Map.class); job.setReducerClass(Reduce.class);
    job.setOutputKeyClass(Text.class); job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0])); FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

Command to run

```

hadoop jar MapReduce\build\flightanalysis.jar flightanalysis.MonthlyFlightCount data\flights.csv
MapReduce\output\03_monthly_flight_count

```

```

E:\BDAL\Final_Assignment\Flight_Data_Analysis>hadoop jar MapReduce\build\flightanalysis.jar flightanalysis.MonthlyFlightCount data\flights.csv MapReduce\output\03_monthly_flight_count
Map input records=5819080
Reduce output records=12
map 100% reduce 100%
Job completed successfully

```

```

E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/MapReduce/03_monthly_flight_count/part-r-00000
1      469968
10     486165
11     467972
12     479230
2      429191
3      504312
4      485151
5      496993
6      503897
7      520718
8      510536
9      464946

```

Selected execution output

Key	Result
January	469,968
March	504,312
July	520,718
August	510,536
December	479,230

4) Cancelled Flights by Airline

Filters cancelled records and counts them for each airline.

Code

```

package flightanalysis;

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;

```

```

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

/** Counts cancelled flights for each airline. */
public class CancellationByAirline {
    public static class Map extends Mapper<LongWritable, Text, Text, IntWritable> {
        private static final IntWritable ONE = new IntWritable(1);
        public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
            String[] f = value.toString().split(",", -1);
            if (f.length == 31 && !"YEAR".equals(f[0]) && "1".equals(f[24]) && !f[4].trim().isEmpty()) context.write(new Text(f[4]), ONE);
        }
    }
    public static class Reduce extends Reducer<Text, IntWritable, Text, IntWritable> {
        public void reduce(Text key, Iterable<IntWritable> values, Context context) throws IOException, InterruptedException {
            int total = 0; for (IntWritable value : values) total += value.get(); context.write(key, new IntWritable(total));
        }
    }
    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration(); conf.set("fs.defaultFS", "file:///"); conf.set("mapreduce.framework.name", "local");
        Job job = Job.getInstance(conf, "Cancelled flights by airline");
        job.setJarByClass(CancellationByAirline.class); job.setMapperClass(Map.class); job.setReducerClass(Reduce.class);
        job.setOutputKeyClass(Text.class); job.setOutputValueClass(IntWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0])); FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}

```

Command to run

```

hadoop jar MapReduce/build\flightanalysis.jar flightanalysis.CancellationByAirline data\flights.csv
MapReduce\output\04_cancellation_by_airline

```

```

E:\BDAL\Final_Assignment\Flight_Data_Analysis>hadoop jar MapReduce\build\flightanalysis.jar flightanalysis.CancellationByAirline data\flights.csv MapReduce\output\04_cancellation_by_airline

Map input records=5819080
Map output records=89884
Reduce output records=14
Job completed successfully

```

```

E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/MapReduce/04_cancellation_by_airline/part-r-00000
AA      10919
AS      669
B6      4276
DL      3824
EV      15231
F9      588
HA      171
MQ      15025
NK      2004
OO      9960
UA      6573
US      4067
VX      534
WN      16043

```

Selected execution output

Key	Result
WN	16,043
EV	15,231
MQ	15,025
AA	10,919
OO	9,960

5) Average Arrival Delay by Airline

Calculates mean recorded arrival delay in minutes for each airline.

Code

```
package flightanalysis;

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

/** Calculates mean recorded arrival delay, in minutes, for each airline. */
public class AverageArrivalDelayByAirline {
    public static class Map extends Mapper<LongWritable, Text, Text, Text> {
        public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
            String[] f = value.toString().split(",", -1);
            if (f.length == 31 && !"YEAR".equals(f[0]) && !f[4].trim().isEmpty() && !f[22].trim().isEmpty()) {
                try { context.write(new Text(f[4]), new Text(f[22] + "\t1")); } catch (NumberFormatException ignored) { }
            }
        }
    }
    public static class Reduce extends Reducer<Text, Text, Text, Text> {
        public void reduce(Text key, Iterable<Text> values, Context context) throws IOException, InterruptedException {
            double sum = 0; long count = 0;
            for (Text value : values) { String[] pair = value.toString().split("\t"); sum += Double.parseDouble(pair[0]); count += Long.parseLong(pair[1]); }
            context.write(key, new Text(String.format("%.2f", (double) sum / count)));
        }
    }
    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration(); conf.set("fs.defaultFS", "file:///"); conf.set("mapreduce.framework.name", "local");
        Job job = Job.getInstance(conf, "Average arrival delay by airline");
        job.setJarByClass(AverageArrivalDelayByAirline.class); job.setMapperClass(Map.class); job.setReducerClass(Reduce.class);
        job.setMapOutputKeyClass(Text.class); job.setMapOutputValueClass(Text.class); job.setOutputKeyClass(Text.class); job.setOutputValueClass(Text.class);
        FileInputFormat.addInputPath(job, new Path(args[0])); FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Command to run

```
hadoop jar MapReduce/build/flightanalysis.jar flightanalysis.AverageArrivalDelayByAirline data/flights.csv
MapReduce/output/05_average_arrival_delay_by_airline
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hadoop jar MapReduce/build/flightanalysis.jar flightanalysis.AverageArrivalDelayByAirline data/flights.csv MapReduce/output/05_average_arrival_delay_by_airline

Map input records=5819888

Map output records=5714088

Reduce output records=14

Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/MapReduce/05_average_arrival_delay_by_airline/part-r-00000
AA      3.45
AS     -0.98
B6      6.68
DL      8.19
EV      6.59
F9     12.50
HA      2.82
MQ      6.46
NK     14.47
OO      5.85
UA      5.43
US      3.71
VX      4.74
WN      4.37
```

Selected execution output

Key	Result
NK	14.47
F9	12.50
B6	6.68
EV	6.59
MQ	6.46

Pig Flight Data Analysis

1) Airline Flight Totals

Groups flights by airline, joins the airline lookup, and orders the totals.

Script

```
flights = LOAD 'E:/BDAL/New_Dataset/flights.csv' USING PigStorage(',') AS
(year:int,month:int,day:int,day_of_week:int,airline:chararray,flight_number:int,tail_number:chararray,origin:chararray,destination:chararray,scheduled_departur
e:chararray,departure_time:chararray,departure_delay:double,taxi_out:double,wheels_off:chararray,scheduled_time:double,elapsed_time:double,air_time:double
,distance:double,wheels_on:chararray,taxi_in:double,scheduled_arrival:chararray,arrival_time:chararray,arrival_delay:double,diverted:int,cancelled:int,cancellati
on_reason:chararray,air_system_delay:double,security_delay:double,airline_delay:double,late_aircraft_delay:double,weather_delay:double);

airlines = LOAD 'E:/BDAL/New_Dataset/airlines.csv' USING PigStorage(',') AS (code:chararray,airline_name:chararray);
valid_flights = FILTER flights BY year IS NOT NULL;
valid_airlines = FILTER airlines BY code != 'IATA_CODE';
grouped = GROUP valid_flights BY airline;
totals = FOREACH grouped GENERATE group AS airline_code, COUNT(valid_flights) AS total_flights;
named = JOIN totals BY airline_code, valid_airlines BY code;

result = FOREACH named GENERATE totals::airline_code AS airline_code, valid_airlines::airline_name AS airline_name, totals::total_flights AS
total_flights;
ordered = ORDER result BY total_flights DESC;

STORE ordered INTO 'E:/BDAL/Report_1/Flight_Data_Analysis/outputs/pig/01_airline_flight_totals' USING PigStorage('');
```

Command to run

```
pig -x local -f PigAnalysis\scripts\01_airline_flight_totals.pig
```

```
Microsoft Windows [Version 10.0.26100.0]
(c) Microsoft Corporation. All rights reserved.
C:\Windows\System32>cd /d E:\BDAL\Final_Assignment\Flight_Data_Analysis

E:\BDAL\Final_Assignment\Flight_Data_Analysis>pig -x local -f PigAnalysis\scripts\01_airline_flight_totals.pig
Pig features: HASH_JOIN, GROUP_BY, ORDER_BY, FILTER
Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/PIG/01_airline_flight_totals/part-r-00000
WN|Southwest Airlines Co.|1261855
DL|Delta Air Lines Inc.|875881
AA|American Airlines Inc.|725984
OO|Skywest Airlines Inc.|588353
EV|Atlantic Southeast Airlines|571977
UA|United Air Lines Inc.|515723
MQ|American Eagle Airlines Inc.|294632
B6|JetBlue Airways|267048
US|US Airways Inc.|198715
AS|Alaska Airlines Inc.|172521
NK|Spirit Air Lines|117379
F9|Frontier Airlines Inc.|90836
HA|Hawaiian Airlines Inc.|76272
VX|Virgin America|61903
```

2) Monthly Flight Totals

Groups valid flights by month and counts them in calendar order.

Script

```
flights = LOAD 'E:/BDAL/New_Dataset/flights.csv' USING PigStorage(',') AS
(year:int,month:int,day:int,day_of_week:int,airline:chararray,flight_number:int,tail_number:chararray,origin:chararray,destination:chararray,scheduled_departur
e:chararray,departure_time:chararray,departure_delay:double,taxi_out:double,wheels_off:chararray,scheduled_time:double,elapsed_time:double,air_time:double
,distance:double,wheels_on:chararray,taxi_in:double,scheduled_arrival:chararray,arrival_time:chararray,arrival_delay:double,diverted:int,cancelled:int,cancellati
on_reason:chararray,air_system_delay:double,security_delay:double,airline_delay:double,late_aircraft_delay:double,weather_delay:double);

valid = FILTER flights BY year IS NOT NULL;
grouped = GROUP valid BY month;
result = FOREACH grouped GENERATE group AS month, COUNT(valid) AS total_flights;
ordered = ORDER result BY month ASC;

STORE ordered INTO 'E:/BDAL/Report_1/Flight_Data_Analysis/outputs/pig/02_monthly_flight_totals' USING PigStorage("");
```

Command to run

```
pig -x local -f PigAnalysis\scripts\02_monthly_flight_totals.pig
```

```
C:\Windows\System32>cd /d E:\BDAL\Final_Assignment\Flight_Data_Analysis

E:\BDAL\Final_Assignment\Flight_Data_Analysis>pig -x local -f PigAnalysis\scripts\02_monthly_flight_totals.pig

Pig features: GROUP_BY, ORDER_BY, FILTER

Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/PIG/02_monthly_flight_totals/part-r-00000
1|469968
2|429191
3|504312
4|485151
5|496993
6|503897
7|520718
8|510536
9|464946
10|486165
11|467972
12|479230
```

3) Top Origin Airports

Counts departures per origin airport, orders them descending, and limits the result to ten.

Script

```
flights = LOAD 'E:/BDAL/New_Dataset/flights.csv' USING PigStorage(',') AS
(year:int,month:int,day:int,day_of_week:int,airline:chararray,flight_number:int,tail_number:chararray,origin:chararray,destination:chararray,scheduled_departure:chararray,departure_time:chararray,departure_delay:double,taxi_out:double,wheels_off:chararray,scheduled_time:double,elapsed_time:double,air_time:double,distance:double,wheels_on:chararray,taxi_in:double,scheduled_arrival:chararray,arrival_time:chararray,arrival_delay:double,diverted:int,cancelled:int,cancellation_reason:chararray,air_system_delay:double,security_delay:double,airline_delay:double,late_aircraft_delay:double,weather_delay:double);

valid = FILTER flights BY year IS NOT NULL AND origin IS NOT NULL;
grouped = GROUP valid BY origin;
counts = FOREACH grouped GENERATE group AS origin_airport, COUNT(valid) AS departures;
ordered = ORDER counts BY departures DESC;

result = LIMIT ordered 10;
STORE result INTO 'E:/BDAL/Report_1/Flight_Data_Analysis/outputs/pig/03_top_origin_airports' USING PigStorage('');
```

Command to run

```
pig -x local -f PigAnalysis\scripts\03_top_origin_airports.pig
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>pig -x local -f PigAnalysis\scripts\03_top_origin_airports.pig
Pig features: GROUP_BY, ORDER_BY, LIMIT, FILTER
Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/PIG/03_top_origin_airports/part-r-00000
ATL|346836
ORD|285884
DFW|239551
DEN|196055
LAX|194673
SFO|148008
PHX|146815
IAH|146622
LAS|133181
MSP|112117
```

4) Top Routes

Groups flights by origin and destination, then returns the ten busiest directional routes.

Script

```
flights = LOAD 'E:/BDAL/New_Dataset/flights.csv' USING PigStorage(',') AS
(year:int,month:int,day:int,day_of_week:int,airline:chararray,flight_number:int,tail_number:chararray,origin:chararray,destination:chararray,scheduled_departure:chararray,departure_time:chararray,departure_delay:double,taxi_out:double,wheels_off:chararray,scheduled_time:double,elapsed_time:double,air_time:double,distance:double,wheels_on:chararray,taxi_in:double,scheduled_arrival:chararray,arrival_time:chararray,arrival_delay:double,diverted:int,cancelled:int,cancellation_reason:chararray,air_system_delay:double,security_delay:double,airline_delay:double,late_aircraft_delay:double,weather_delay:double);

valid = FILTER flights BY year IS NOT NULL AND origin IS NOT NULL AND destination IS NOT NULL;
grouped = GROUP valid BY (origin, destination);
counts = FOREACH grouped GENERATE FLATTEN(group) AS (origin_airport, destination_airport), COUNT(valid) AS total_flights;
ordered = ORDER counts BY total_flights DESC;

result = LIMIT ordered 10;
STORE result INTO 'E:/BDAL/Report_1/Flight_Data_Analysis/outputs/pig/04_top_routes' USING PigStorage('');
```

Command to run

```
pig -x local -f PigAnalysis\scripts\04_top_routes.pig
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>pig -x local -f PigAnalysis\scripts\04_top_routes.pig
Pig features: GROUP_BY, FOREACH, FLATTEN, ORDER_BY, LIMIT
Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/PIG/04_top_routes/part-r-00000
SFO|LAX|13744
LAX|SFO|13457
JFK|LAX|12016
LAX|JFK|12015
LAS|LAX|9715
LGA|ORD|9639
LAX|LAS|9594
ORD|LGA|9575
SFO|JFK|8440
JFK|SFO|8437
```

5) Average Departure Delay by Airline

Filters records with a departure delay and calculates airline-level averages.

Script

```
flights = LOAD 'E:/BDAL/New_Dataset/flights.csv' USING PigStorage(',') AS
(year:int,month:int,day:int,day_of_week:int,airline:chararray,flight_number:int,tail_number:chararray,origin:chararray,destination:chararray,scheduled_departure:chararray,departure_time:chararray,departure_delay:double,taxi_out:double,wheels_off:chararray,scheduled_time:double,elapsed_time:double,air_time:double,distance:double,wheels_on:chararray,taxi_in:double,scheduled_arrival:chararray,arrival_time:chararray,arrival_delay:double,diverted:int,cancelled:int,cancellation_reason:chararray,air_system_delay:double,security_delay:double,airline_delay:double,late_aircraft_delay:double,weather_delay:double);

valid = FILTER flights BY year IS NOT NULL AND departure_delay IS NOT NULL;
grouped = GROUP valid BY airline;
result = FOREACH grouped GENERATE group AS airline_code, AVG(valid.departure_delay) AS average_departure_delay_minutes;
ordered = ORDER result BY average_departure_delay_minutes DESC;

STORE ordered INTO 'E:/BDAL/Report_1/Flight_Data_Analysis/outputs/pig/05_average_departure_delay_by_airline' USING PigStorage('|');
```

Command to run

```
pig -x local -f PigAnalysis\scripts\05_average_departure_delay_by_airline.pig
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>pig -x local -f PigAnalysis\scripts\05_average_departure_delay_by_airline.pig
Pig features: GROUP_BY, AVG, ORDER_BY, FILTER
Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/PIG/05_average_departure_delay_by_airline/part-r-00000
NK|15.944765880783688
UA|14.435441010805953
F9|13.350858345331709
B6|11.5143526744102
WN|10.581986295158847
MQ|10.125188203309524
VX|9.022595096521952
AA|8.908856346719806
EV|8.71593449776958
OO|7.801103808415331
DL|7.36925417661782
US|6.141136917746696
AS|1.7858007096736666
HA|0.48571315965790407
```

6) Cancellations by Airline

Filters cancelled flights, groups them by airline, and orders the cancellation counts.

Script

```
flights = LOAD 'E:/BDAL/New_Dataset/flights.csv' USING PigStorage(',') AS
(year:int,month:int,day:int,day_of_week:int,airline:chararray,flight_number:int,tail_number:chararray,origin:chararray,destination:chararray,scheduled_departure:chararray,departure_time:chararray,departure_delay:double,taxi_out:double,wheels_off:chararray,scheduled_time:double,elapsed_time:double,air_time:double,distance:double,wheels_on:chararray,taxi_in:double,scheduled_arrival:chararray,arrival_time:chararray,arrival_delay:double,diverted:int,cancelled:int,cancellation_reason:chararray,air_system_delay:double,security_delay:double,airline_delay:double,late_aircraft_delay:double,weather_delay:double);

cancelled_flights = FILTER flights BY year IS NOT NULL AND cancelled == 1;
grouped = GROUP cancelled_flights BY airline;
result = FOREACH grouped GENERATE group AS airline_code, COUNT(cancelled_flights) AS cancelled_flights;
ordered = ORDER result BY cancelled_flights DESC;

STORE ordered INTO 'E:/BDAL/Report_1/Flight_Data_Analysis/outputs/pig/06_cancellations_by_airline' USING PigStorage('|');
```

Command to run

```
pig -x local -f PigAnalysis\scripts\06_cancellations_by_airline.pig
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>pig -x local -f PigAnalysis\scripts\06_cancellations_by_airline.pig
Pig features: FILTER, GROUP_BY, ORDER_BY
Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/PIG/06_cancellations_by_airline/part-r-00000
WN|16043
EV|15231
MQ|15025
AA|10919
OO|9960
UA|6573
B6|4276
US|4067
DL|3824
NK|2004
AS|669
F9|588
VX|534
HA|171
```

7) Average Arrival Delay by Month

Computes the average arrival delay for each month.

Script

```
flights = LOAD 'E:/BDAL/New_Dataset/flights.csv' USING PigStorage(',') AS
(year:int,month:int,day:int,day_of_week:int,airline:chararray,flight_number:int,tail_number:chararray,origin:chararray,destination:chararray,scheduled_departure:chararray,departure_time:chararray,departure_delay:double,taxi_out:double,wheels_off:chararray,scheduled_time:double,elapsed_time:double,air_time:double,distance:double,wheels_on:chararray,taxi_in:double,scheduled_arrival:chararray,arrival_time:chararray,arrival_delay:double,diverted:int,cancelled:int,cancellation_reason:chararray,air_system_delay:double,security_delay:double,airline_delay:double,late_aircraft_delay:double,weather_delay:double);

valid = FILTER flights BY year IS NOT NULL AND arrival_delay IS NOT NULL;
grouped = GROUP valid BY month;
result = FOREACH grouped GENERATE group AS month, AVG(valid.arrival_delay) AS average_arrival_delay_minutes;
ordered = ORDER result BY month ASC;

STORE ordered INTO 'E:/BDAL/Report_1/Flight_Data_Analysis/outputs/pig/07_average_arrival_delay_by_month' USING PigStorage('|');
```

Command to run

```
pig -x local -f PigAnalysis\scripts\07_average_arrival_delay_by_month.pig
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>pig -x local -f PigAnalysis\scripts\07_average_arrival_delay_by_month.pig
Pig features: FILTER, GROUP_BY, AVG, ORDER_BY
Job completed successfully
```

```
E:\BDAL\Final_Assignment\Flight_Data_Analysis>hdfs dfs -cat /Final_Project/PIG/07_average_arrival_delay_by_month/part-r-00000
1|5.813582983416227
2|8.320500020850556
3|4.92067265685641
4|3.1631900611579318
5|4.485018615679651
6|9.601590351569554
7|6.4317747052785466
8|4.607372469025074
9|-0.7725709883956179
10|-0.780389663641748
11|1.100783576682592
12|6.092902747824754
```

Conclusion

The full dataset contains 5,819,079 flight records. Southwest Airlines operated the most flights (1,261,855). ATL was the busiest origin airport with 346,836 departures. July had the highest monthly flight total at 520,718. The busiest directional route was SFO to LAX with 13,744 flights. The highest average departure delay was recorded by NK at approximately 15.94 minutes, while the highest average arrival delay was also NK at 14.47 minutes. These findings are derived directly from the executed MapReduce and Pig outputs.